

L4YAML: A Verified YAML 1.2.2 Parser in Lean 4

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L4YAML is a fully verified YAML 1.2.2 parser written in pure Lean 4. It delivers 2,309 machine-checked theorems across 61 proof modules, zero axioms, zero `sorry`, and zero `partial def` — while passing 100% of the YAML 1.2.2 specification examples and 100% of the applicable `yaml-test-suite` test IDs (225/225).

This manual documents the project's architecture, verification strategy, security model, and FFI bindings for C, Python, and Rust.

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- 2. Architecture
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2. Architecture

L4YAML implements a two-pass pipeline that mirrors the layered structure of the YAML 1.2.2 specification itself. Input bytes flow through a character-level [scanner](#), producing a typed token stream, which a [recursive-descent parser](#) converts into a structured AST.

2.1. Scanner (Lexical Layer)

2.2. Token Parser (Syntactic Layer)

2.3. Type System

2.4. Module Organization

2.5. Import Graph

2.5. Import Graph

The project's dependency structure can be visualized using `lake exe graph`. The FFI layer sits at the top, depending on `Config`, which in turn depends on the scanner/parser/schema pipeline:

```
FFI ← Config ← Dump ← Schema ← Token ← Scanner ← TokenParser
      ↑
CharPredicates ← YamlSpec
```

The full import graph and component-level dependency graphs are available in the `graphs/` directory as Graphviz DOT files.

2.4. Module Organization

The project is organized into several module groups:

Group	Key Modules	Purpose
Core	Types.lean, Token.lean, Grammar.lean, YamlSpec.lean, CharPredicates.lean	Type definitions, token types, grammar inductive, spec production predicates
Scanner	Scanner.lean, Scanner/Whitespace.lean, Scanner/Scalar.lean, ...	Character-to-token conversion with full state management
Parser	TokenParser.lean, NodeParser.lean	Token-to-AST recursive descent
Validation	Limits.lean, Schema.lean	Security limits, Core Schema type resolution
Output	Emitter.lean, Dump.lean, RoundTrip.lean	Canonical emitter (~164L), style-aware dump, round-trip properties
FFI	FFI.lean, ffi/, python/, rust/	C/Python/Rust bindings via @[export]
Proofs	Proofs/ (61 modules, ~47,000 lines)	Machine-checked theorems for soundness, completeness, progress, well-formedness

2.1. Scanner (Lexical Layer)

The scanner (`Scanner.lean`, ~920 lines) converts a UTF-8 input string into a stream of `YamlToken` values. It implements 132 YAML productions from the specification and maintains several pieces of state:

- *Indentation stack* — tracks block-level nesting via column positions
- *Flow level counter* — distinguishes block context from flow context (`[ ]`, `{ }`)
- *Simple key state* — manages the YAML "simple key" lifecycle where a plain scalar may retroactively become a mapping key when `:` is encountered
- *Anchor map* — records anchor names for alias resolution
- *Position tracking* — offset, line, and column for every token

The scanner returns `Except ScanError (Array YamlToken)`, where `ScanError` carries position information and a human-readable message.

2.1.1. Token Types

The `YamlToken` inductive defines the full set of lexical elements:

- Stream markers: `streamStart`, `streamEnd`
- Document markers: `documentStart`, `documentEnd`
- Structure indicators: `blockSequenceStart`, `blockMappingStart`, `blockEnd`, `flowSequenceStart`, `flowSequenceEnd`, `flowMappingStart`, `flowMappingEnd`, `key`, `value`, `blockEntry`, `flowEntry`
- Content tokens: `scalar` (with style tag: `plain`, `single-quoted`, `double-quoted`, `literal`, `folded`), `anchor`, `alias`, `tag`
- Placeholder tokens: used as reservation slots for simple key backpatching (filtered before output)

2.1.2. Append-Only Design

A critical design choice is the *append-only token stream*. When the scanner encounters a potential simple key, it pushes two placeholder tokens (for the future key and `blockMappingStart` indicators) and records their indices. If the key is later confirmed (by encountering `:`), these placeholders are overwritten in-place via `setIfInBounds`. If not confirmed, they remain as placeholders and are filtered out before the token stream is returned.

This avoids `insertAt` operations that would shift array indices and invalidate saved positions — a property that is essential for the formal proof that the scanner makes monotonic progress.

2.2. Token Parser (Syntactic Layer)

The token parser (`TokenParser.lean`, ~426 lines) consumes the `YamlToken` stream and produces an `Array YamlDocument`. It implements 54 YAML productions via hand-written recursive descent (no parser combinator library).

Key responsibilities:

- *Multi-document support* — handles `---` and `...` boundaries
- *Alias resolution* — substitutes `*anchor` references with previously anchored values
- *Tag resolution* — applies `%TAG` directive handle expansion and schema-level type resolution
- *Schema application* — resolves untagged scalars to typed values (`null`, `bool`, `int`, `float`, `str`) via the Core Schema

2.3. Type System

The output AST is centered on `YamlValue`, a compact inductive type:

- `YamlValue.scalar` — tagged string value with resolved `YamlType`
- `YamlValue.sequence` — ordered list of values
- `YamlValue.mapping` — list of key-value pairs (preserving order)

Each value is wrapped in `YamlDocument`, which carries document-level metadata including directive tags and version indicators.

Position tracking is provided by `YamlPos` (offset, line, column), enabling precise error reporting that maps back to the original input.

5. FFI Bindings

L4YAML provides foreign function interface bindings for C, Python, and Rust, enabling the verified parser to be used from mainstream languages. The FFI layer sits above the full parser pipeline — callers get the same verified parsing logic, security limits, and schema resolution available in Lean.

5.1. Dependency Graph

5.2. C API

5.3. Python API

5.4. Rust API

5.5. Adding FFI Bindings

5.5. Adding FFI Bindings

To expose a new Lean function via FFI:

1. Add `@[export l4yaml_function_name]` to the Lean definition
2. Declare the corresponding C prototype in `ffi/l4yaml.h`
3. Add any necessary shim code in `ffi/l4yaml_shim.c`
4. Update the Python/Rust wrappers as needed
5. Add tests at each layer

The `@[export]` attribute instructs the Lean compiler to generate a C-callable symbol with the specified name.

5.2. C API

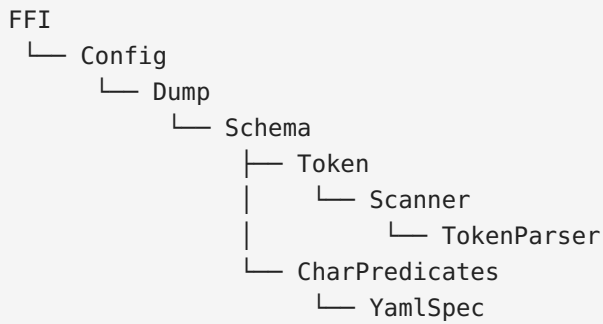
The C API is defined in `ffi/l4yaml.h` with a bridge layer in `ffi/l4yaml_shim.c`. It provides approximately 26 exported functions covering:

- *Parsing* — `l4yaml_parse_safe` and related functions
- *Dumping* — converting `YamlValue` back to YAML text
- *Schema* — tag handle registration and type resolution
- *Limits* — configuring `ParserLimits` via preset codes or individual parameter setters
- *Memory* — proper Lean object lifecycle management (`lean_inc_ref` / `lean_dec_ref`)

All C API functions operate on opaque Lean object pointers. The shim layer handles the Lean runtime initialization and provides stable C-callable entry points.

5.1. Dependency Graph

The FFI module sits at the top of the dependency chain:



`FFI.lean` exports functions via `@[export]` that are callable from C. The Python and Rust bindings wrap these C-level exports with idiomatic APIs in their respective languages.

5.3. Python API

The Python package (`python/l4yaml/`) wraps the C API using `ctypes`, providing a Pythonic interface with:

- Full type annotations for IDE support
- Automatic garbage collection of Lean objects via reference counting
- Cross-platform support (Linux, macOS, Windows)
- 78 tests covering the full API surface

Usage follows the familiar pattern:

```
from l4yaml import parse

result = parse("key: value")
```


5.4. Rust API

The Rust workspace (`rust/`) provides two crates:

- **`l4yam1-sys`** — raw FFI bindings generated from the C header. Unsafe, low-level, intended for direct interop.
- **`l4yam1`** — safe RAI wrapper with 21 tests. Lean object lifetimes are managed via the `Drop` trait, ensuring proper cleanup without manual reference counting.

The safe wrapper provides idiomatic Rust types and error handling via `Result`, translating Lean's `Except` into Rust's error model.

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1. Overview

L4YAML is a pure Lean 4 implementation of the YAML 1.2.2 specification (RFC 9512). No external parsing libraries, no C dependencies in the core, and no use of `partial def` — every function is provably terminating.

The project demonstrates that a production-quality parser for a complex real-world format can be built and formally verified in Lean 4, with practical performance and comprehensive test coverage.

1.1. At a Glance

1.2. Key Design Decisions

1.1. At a Glance

Key Metric	Value
Machine-checked theorems	2,309 across 61 proof modules (~47,000 lines)
Compile-time #guard tests	2,124 (kernel-evaluated at build time)
Axioms / sorry / partial def	Zero / Zero / Zero
Build jobs	415/415 (clean, zero warnings)
Runtime test suites	1,041 tests across 19 suites
Spec examples passing	132/132 (100%)
yaml-test-suite IDs	225/225 YAML 1.2.2-applicable (100%)
yaml-test-suite total	354/406 (87.2%; 52 skipped are YAML 1.1/1.3)

1.2. Key Design Decisions

The parser is built around several deliberate design choices:

- **Pure Lean 4, zero external dependencies.** The core parser has no C FFI calls and no dependency on external parsing libraries. This ensures every line of parsing logic is visible to the Lean kernel for formal verification.
- **Total functions only.** Every function terminates provably. The `partial` keyword is never used in production code. Termination is proved via well-founded recursion over input offsets, indentation levels, and flow nesting depth.
- **Append-only token streams.** Tokens are emitted into a pre-allocated array with placeholder reservation slots. Backpatching uses `setIfInBounds` (bounded update) rather than `insertAt` (which would shift indices). This design ensures monotonic progress and simplifies the formal proof of scanner correctness.
- **Two-pass architecture.** A character-level scanner emits a `YamlToken` stream, which a separate recursive-descent parser converts to a typed AST. This separation mirrors the YAML specification's own layered structure and enables independent verification of each layer.
- **Configurable security limits.** All resource bounds (nesting depth, string length, collection sizes) are configurable via `ParserLimits`, with four built-in presets from `strict` to `unlimited`.

4. Security

YAML parsing is a well-known attack surface. L4YAML addresses this with configurable parser limits, preset security profiles, and formal verification of the parsing logic itself.

4.1. Threat Model

4.2. ParserLimits

4.3. Preset Configurations

4.4. Verification of Security Properties

4.2. ParserLimits

The ParserLimits structure provides 11 configurable threat mitigations:

Parameter	Default	Purpose
nestingDepth	250	Maximum recursion depth — prevents billion-laugh expansion
maxLength	10 MB	Maximum scalar string length — DoS prevention
maxArrayLength	100,000	Maximum sequence element count
maxObjectSize	10 MB	Maximum total mapping size
maxAliasDepth	50	Maximum alias chain depth — recursive cycle protection
allowDuplicateKeys	false	Whether duplicate mapping keys are accepted
allowedTagHandles	customizable	Restricts which tag handles (!, !!, custom) are permitted
forbiddenTags	customizable	Explicit rejection of dangerous tags (e.g., !!python/object)
parseErrorPolicy	strict	Whether non-conformant input is rejected or best-effort parsed
commentEncoding	explicit	Character encoding validation for comments
literalNewlineHandling	standard	Newline normalization in literal scalars

4.3. Preset Configurations

Four preset configurations are provided for common use cases:

- **strict** — all protections enabled at conservative thresholds. Recommended for processing untrusted input (e.g., user uploads, network-received configuration).
- **default** — balanced settings suitable for most applications. Limits are generous enough for typical configuration files while still preventing resource exhaustion.
- **permissive** — reduced validation for trusted input. Useful when parsing known-good YAML from controlled sources.
- **unlimited** — all limits disabled. Explicitly dangerous; intended only for testing or for processing input that has already been validated externally.

The FFI layer exposes presets via `presetToLimits`, which maps a `UInt8` preset code to the corresponding `ParserLimits` configuration.

4.1. Threat Model

YAML parsers face several categories of attack:

- *Billion-laugh attacks* — deeply nested aliases that expand exponentially, exhausting memory
- *Denial of service* — extremely long strings, deeply nested structures, or very large collections that consume excessive resources
- *Arbitrary code execution* — YAML tags that trigger object deserialization in languages with unsafe constructors (e.g., Python's `!!python/object`)
- *Duplicate key confusion* — multiple identical keys in a mapping, where different consumers may pick different values

L4YAML mitigates all of these through its `ParserLimits` configuration.

4.4. Verification of Security Properties

The formal proofs establish that the parser correctly enforces its configured limits:

- Nesting depth is checked on every recursive call
- String length is checked during scalar accumulation
- Collection sizes are checked during sequence/mapping construction
- Alias depth is bounded during resolution

Because the parser is written in pure Lean 4 with no unsafe FFI in the core, there is no possibility of buffer overflows, use-after-free, or other memory safety violations in the parsing logic. The Lean runtime provides automatic memory management via reference-counted garbage collection.

7. Test Results

The L4YAML CI pipeline runs every test suite on each push and pull request. This chapter describes the test result categories and links to the most recent CI-generated reports.

Test results are generated by the `test-coverage.yml` GitHub Actions workflow and published alongside this documentation in the `reports/` directory.

The interactive coverage dashboard is available at reports/index.html.

7.1. yaml-test-suite Coverage

7.2. Lean Test Suites

7.3. Diagnostic Checks

7.4. FFI Test Results

7.3. Diagnostic Checks

Diagnostic pipelines produce detailed output for regression tracking:

Diagnostic	Output File	Description
Flow Regression	flowregressioncheck.txt	Flow parsing edge case regression tracking
Error Stage	errorstagediag.txt	Error stage diagnostic output
Scalar Stage	scalarstagediag.txt	Scalar stage diagnostic output

7.4. FFI Test Results

The FFI layer is tested through three independent paths:

7.4.1. Python FFI Tests

Two Python test suites validate the ctypes-based bindings:

- *FFI Integration Tests* (`Tests/test_python_ffi.py`) — tests the C shared library directly from Python
- *Package Tests* (`python/tests/`) — tests the `l4yaml` Python package API

Results are captured as both text logs and JUnit XML, with an HTML report generated from the JUnit data.

7.4.2. Cross-Language Comparison

The cross-language comparison runs the full `yaml-test-suite` through all four backends (Lean, C, Python, Rust) across five limit presets (`unlimited`, `default`, `strict`, `permissive`, `safe_tags`).

This verifies that the FFI layer preserves identical parsing behavior — every backend must produce the same `pass/fail/skip` results for every test case under every preset configuration.

Results are available at [Cross-Language Comparison Report](#).

7.2. Lean Test Suites

The following Lean-native test suites are executed and their output captured:

Suite	Output File	Description
Unit Tests	unit-tests.txt	Core unit tests for parser internals
Demo	demo.txt	End-to-end parsing demonstrations
Explicit Key Tests	explicitkeytests.txt	YAML explicit key (?) handling
Flow Tests	flowtests.txt	Flow collection syntax completeness
Validation Tests	validationtests.txt	Input validation and error reporting
Dump Round-Trip	dumproundtrip.txt	Parse → dump → parse cycle verification
Raw Parse Tests	rawparsetests.txt	Token-to-AST parsing validation
Spec Examples	specexamples.txt	All 132 YAML 1.2.2 specification examples
Schema Dump	schemadump.txt	Schema resolution and dump formatting
Scanner Tests	scannertests.txt	Scanner behavior on targeted inputs
Scanner Spec Examples	scannerspecexamples.txt	Scanner-level tokens for spec examples
Adversarial Tests	adversarialtests.txt	Handcrafted adversarial inputs
Mutation Tests	mutationtests.txt	Systematic input perturbation
Property Tests	propertytests.txt	Randomized property-based testing
Production Coverage	productioncoverage.txt	Coverage across production-style inputs
Limit Tests	limittests.txt	Parser security limit enforcement

7.1. yaml-test-suite Coverage

The `suiterunner` executable runs all 406 unique test cases from the official `yaml-test-suite` and generates interactive HTML coverage reports with filtering, sorting, and per-stage breakdown.

Interactive reports:

- [Full Coverage Report](#) — all test cases with filtering and sorting
- [Verified Tests Detail](#) — all internal test suites with per-test results
- [Production Coverage](#) — YAML 1.2.2 productions cross-referenced with `@[yaml_spec]` annotations

Coverage by stage:

- [Scalar Tests](#)
- [Flow Tests](#)
- [Block Tests](#)
- [Document Tests](#)
- [Advanced Tests](#)
- [Error Tests](#)

6. Testing

Beyond the 2,309 formal theorems, L4YAML maintains extensive runtime and compile-time test suites that validate the parser against real-world YAML inputs and the official specification.

6.1. Specification Coverage

6.2. Test Suites

6.3. Running Tests

6.3. Running Tests

All tests are built as Lake executables. To run the full default test suite:

```
lake build
```

This builds all 415 jobs (zero warnings) including all test executables listed in `defaultTargets`. Individual suites can be run directly:

```
lake exe specexamples  
lake exe scannertests  
lake exe adversarialtests
```

Test results are captured in `docs/` as both plain text and HTML reports.

6.1. Specification Coverage

- **132/132** YAML 1.2.2 specification examples pass (100%)
- **225/225** unique YAML test IDs from the yaml-test-suite pass (100% of YAML 1.2.2-applicable tests)
- **354/406** total yaml-test-suite tests pass (87.2%)
- The 52 skipped tests cover YAML 1.1 and YAML 1.3 features that are outside the YAML 1.2.2 scope

6.2. Test Suites

The project includes 24 test executables spanning 19 hand-written suites and 7 diagnostic pipelines:

Suite	Focus
specexamples	All 132 YAML 1.2.2 specification examples
scannerspecexamples	Scanner-level token stream for spec examples
scannertests	Scanner behavior on targeted inputs
rawparsetests	Raw token-to-AST parsing
flowtests	Flow collection syntax (inline sequences, mappings)
flowregressioncheck	Regression tests for flow parsing edge cases
explicitkeytests	Explicit key (?) handling
validationtests	Input validation and error reporting
limittests	Parser limit enforcement
adversarialtests	Adversarial inputs targeting parser robustness
mutationtests	Systematic input perturbation
propertytests	Randomized property-based testing
dumproundtrip	Parse → dump → parse cycle validation
schemadump	Schema resolution and dump formatting
productioncoverage	Coverage analysis across production inputs
errorstagediag	Error stage diagnostic output
scalarestagediag	Scalar stage diagnostic output

3. Verification

L4YAML employs a three-layer verification strategy that combines formal proofs, compile-time checks, and runtime tests to achieve comprehensive coverage of the YAML 1.2.2 specification.

3.1. Three-Layer Strategy

3.2. Key Theorems

3.3. Key Proof Modules

3.4. Theorem Dependency Visualization

3.5. Proof Engineering Patterns

3.6. Zero-Axiom Policy

3.3. Key Proof Modules

Module	Theorems	Scope
ScannerCorrectness.lean	259 theorems + 1,063 guards	Character-to-token correctness for all scanner operations
Completeness.lean	63 theorems	Valid YAML inputs are accepted
Soundness.lean	28 theorems	Output corresponds to valid grammar derivations
RoundTrip.lean	58 theorems + 63 guards	Parse-emit-parse cycle properties
Composition.lean	12 theorems	Scanner + parser pipeline correctness
ScannerEmitBridge.lean	12 theorems + 64 guards	Bridge between scanner emissions and grammar predicates
ParserSoundness.lean	12 theorems	Grammar-to-implementation correspondence
ParserWfaProofs.lean	50 theorems	Well-formed anchors through entire parser pipeline
ParserNodeProofs.lean	57 theorems	Anchor growth and alias resolution
ParserWellBehaved.lean	102 theorems	Flow nesting, token preservation, scannable properties
ScannerProgress.lean	24 theorems	Offset strictly increases on every scanner step
ScannerSimpleKey.lean	Multiple theorems	Simple key lifecycle well-formedness
ScannerDispatch.lean	Multiple theorems	Dispatch pipeline preserves all invariants

3.2. Key Theorems

The following capstone theorems represent the main formal guarantees established by L4YAML. Each is machine-checked by the Lean kernel with zero axioms beyond the built-in foundations.

Each theorem is accompanied by a bipartite dependency graph showing the implementation functions it proves properties about (blue, left) and the supporting theorems used in its proof (green, right). Stdlib lemmas surface in orange when they are domain-relevant.

3.2.1. Pipeline Composition

These theorems establish that the scanner and parser compose correctly into a single end-to-end pipeline.

Theorem	Module	Statement
<code>parseYaml_pipeline</code>	Composition	End-to-end: scan then parse composes correctly. If <code>scanFiltered</code> produces tokens and <code>parseStream</code> accepts them, then <code>parseYaml</code> succeeds with the same result.
<code>parseYamlRaw_pipeline</code>	Composition	Raw pipeline variant without schema resolution.
<code>parseYamlRaw_ok_decompose</code>	Composition	Every successful <code>parseYamlRaw</code> result decomposes into a successful scan step followed by a successful parse step.
<code>parseYaml_ok_iff</code>	Completeness	<code>parseYaml</code> succeeds if and only if the input is valid YAML — the bridge between the implementation and the specification.

3.2.1.1. `parseYaml_pipeline`

`parseYaml_pipeline` dependency graph

3.2.1.2. `parseYamlRaw_pipeline`

`parseYamlRaw_pipeline` dependency graph

3.2.1.3. `parseYamlRaw_ok_decompose`

`parseYamlRaw_ok_decompose` dependency graph

3.2.1.4. `parseYaml_ok_iff`

`parseYaml_ok_iff` dependency graph

3.2.2. Scanner Correctness

Properties of the character-to-token scanner:

Theorem	Module	Statement
<code>scan_produces_valid_tokens</code>	<code>ScannerCorrectness</code>	The scanner output satisfies <code>ValidTokenStream</code> : every emitted token is well-formed, positions are monotonically increasing, and the stream is bracketed by <code>STREAM_START/STREAM_END</code> .
<code>advance_offset_lt</code>	<code>ScannerProgress</code>	Scanner advance <i>strictly</i> increases the byte offset when the offset is within bounds — this is the core termination lemma.
<code>scanLoop_success_emits_streamEnd</code>	<code>ScannerCorrectness</code>	A successful scan loop always terminates with a <code>STREAM_END</code> token.

3.2.2.1. `scan_produces_valid_tokens`

`scan_produces_valid_tokens` dependency graph

3.2.2.2. `advance_offset_lt`

`advance_offset_lt` dependency graph

3.2.2.3. `scanLoop_success_emits_streamEnd`

`scanLoop_success_emits_streamEnd` dependency graph

3.2.3. Parser Correctness

Properties of the token-to-AST parser:

Theorem	Module	Statement
<code>parseStream_sound</code>	<code>ParserSoundness</code>	If the parser produces an AST, it corresponds to a valid YAML grammar derivation.
<code>parseNode_anchors_grow</code>	<code>ParserNodeProofs</code>	The anchor set grows monotonically through <code>parseNode</code> — anchors are never lost during parsing.
<code>parseNode_aliases_resolve'</code>	<code>ParserNodeProofs</code>	Every alias (<code>*name</code>) in the output of <code>parseNode</code> resolves to a previously defined anchor (<code>&name</code>).
<code>parseStream_output_anchors_wellformed</code>	<code>ParserWfaProofs</code>	After <code>parseStream</code> completes, all output anchors are well-formed: every alias target exists and every anchor body is <code>Grammable</code> .

3.2.3.1. `parseStream_sound`

`parseStream_sound` dependency graph

3.2.3.2. `parseNode_anchors_grow`

`parseNode_anchors_grow` dependency graph

3.2.3.3. `parseNode_aliases_resolve'`

`parseNode_aliases_resolve'` dependency graph

3.2.3.4. `parseStream_output_anchors_wellformed`

`parseStream_output_anchors_wellformed` dependency graph

3.2.4. Soundness

Theorems establishing that the AST-to-value conversion faithfully implements the YAML specification:

Theorem	Module	Statement
<code>toYamlValue_correct</code>	Soundness	The <code>toYamlValue</code> function correctly implements the specification's construction rules.
<code>nodeToValue_total</code>	Soundness	Every well-formed AST node can be converted to a <code>YamlValue</code> — the conversion is total.
<code>nodeToValue_deterministic</code>	Soundness	AST-to-value conversion is deterministic: the same input always produces the same output.
<code>scalar_content_preserved</code>	Soundness	Scalar string content is preserved through the parsing pipeline — no characters are added, dropped, or reordered.

3.2.4.1. `toYamlValue_correct`

`toYamlValue_correct` dependency graph

3.2.4.2. `nodeToValue_total`

`nodeToValue_total` dependency graph

3.2.4.3. `nodeToValue_deterministic`

`nodeToValue_deterministic` dependency graph

3.2.4.4. `scalar_content_preserved`

`scalar_content_preserved` dependency graph

3.2.5. Round-Trip Properties

Theorems about the `parse` → `emit` → `parse` cycle:

Theorem	Module	Statement
<code>contentEq_refl</code>	RoundTrip	Content equality is reflexive: every YAML value is content-equal to itself.
<code>contentEq_symm</code>	RoundTrip	Content equality is symmetric.
<code>contentEq_trans</code>	RoundTrip	Content equality is transitive — together with reflexivity and symmetry, this makes <code>contentEq</code> an equivalence relation.
<code>emit_content_invariant</code>	ScannerEmitBridge	The emitter preserves content equality: if two values are content-equal, their emitted canonical forms are identical.
<code>escapeTag_roundtrip</code>	RoundTrip	Tag URI escape and unescape are inverse operations.

3.2.5.1. `contentEq_refl`

`contentEq_refl` dependency graph

3.2.5.2. `contentEq_symm`

`contentEq_symm` dependency graph

3.2.5.3. `contentEq_trans`

`contentEq_trans` dependency graph

3.2.5.4. `emit_content_invariant`

`emit_content_invariant` dependency graph

3.2.5.5. `escapeTag_roundtrip`

`escapeTag_roundtrip` dependency graph

3.5. Proof Engineering Patterns

Several patterns emerged during the verification effort:

- *Decomposition* — large functions are decomposed into validation (error guards), state transformation (pure updates), and emission (token output) phases, each proved independently then composed.
- *Append-only invariant* — the switch from `insertAt` to placeholder reservation slots with `setIfInBounds` backpatching eliminated the hardest class of proof obligations (index shifting).
- *Monotonic progress* — proving `offset_lt` (strict increase) for every scanner operation provides termination and guarantees no infinite loops.
- *Well-formedness threading* — a `WellFormed` predicate on scanner state is threaded through every operation, establishing that invariants are maintained from `scannerInit` through `scanNextToken` to stream completion.
- *Anchor monotonicity* — the `AnchorGrow` relation is proved transitively across all 14 mutually recursive parser functions, establishing that anchors accumulate but are never dropped.
- *Fuel-based termination* — the parser's 14 mutual functions use `fuel : Nat` as a decreasing argument. Initial fuel is set to `4 * tokens.size + 4`, large enough for any valid input.

3.4. Theorem Dependency Visualization

L4YAML includes tooling for visualizing theorem dependencies as bipartite graphs. Each graph shows a key theorem at the center, with:

- *Functions* (left, blue) — the implementation functions the theorem proves a property about
- *Proof dependencies* (right, green) — the supporting theorems used in the proof

3.4.1. Generating Graphs

Generate DOT files for all key theorems:

```
lake build theoremgraph
lake exe theoremgraph tmp/graphs
```

Render to SVG with Graphviz:

```
for f in tmp/graphs/*.dot; do
  dot -Tsvg "$f" -o "${f%.dot}.svg"
done
```

Then open `tmp/graphs/index.html` for an overview page.

Generate a single theorem's graph:

```
lake exe theoremgraph --dot parseYaml_pipeline
```

List all available key theorems:

```
lake exe theoremgraph --list
```

3.4.2. Dependency Extraction

The lower-level depgraph tool extracts three complete dependency graphs from the L4YAML environment:

1. **Function call graph** — definition $\rightarrow$ definition calls
2. **Theorem dependency graph** — theorem $\rightarrow$ theorems used in proof
3. **Function–theorem map** — which functions each theorem is about

```
lake build depgraph
lake exe depgraph --dot calls > calls.dot
lake exe depgraph --dot thmdeps > thmdeps.dot
lake exe depgraph --dot about > about.dot
lake exe depgraph --stats
```

3.4.3. Theorem Coverage Analysis

The analyzethms tool identifies leaf theorems (proved but never cited in other proofs) and native_decide usage patterns:

```
lake build analyzethms
lake exe analyzethms tmp/analysis
```

This writes leaf_thms.json, native_decide_leaves.json, duplicates.json, and stats.txt.

3.1. Three-Layer Strategy

3.1.1. Layer 1: Machine-Checked Proofs

The core layer consists of 2,309 Lean 4 theorems across 61 proof modules (~47,000 lines). These proofs are checked by the Lean kernel — the small trusted core of the system — and establish properties including:

- **Soundness** — every token stream produced by the scanner corresponds to a valid YAML grammar derivation
- **Completeness** — every valid YAML input is accepted (not rejected with an error)
- **Progress** — the scanner's input offset strictly increases on every step, guaranteeing termination
- **Well-formedness preservation** — internal invariants (indentation stack consistency, flow level balance, simple key lifecycle) are maintained across all scanner operations
- **Pipeline composition** — scanner and parser compose correctly to deliver end-to-end guarantees

3.1.2. Layer 2: Compile-Time Guards

2,124 `#guard` statements are evaluated by the Lean kernel at build time. These are not runtime tests — they are *kernel-evaluated assertions* that must hold for the project to compile. A failing `#guard` is a build error, not a test failure.

Guards are used extensively for:

- Concrete scanner behavior on specific inputs
- Round-trip properties (parse → emit → parse = original)
- Token stream structure for specification examples
- Character predicate boundary conditions

3.1.3. Layer 3: Runtime Tests

1,041 runtime tests across 19 suites provide additional coverage:

- *Specification examples* — all 132/132 YAML 1.2.2 examples

- *yaml-test-suite* — 225/225 applicable test IDs (354/406 total; 52 YAML 1.1/1.3 tests are correctly skipped)
- *Property tests* — randomized input generation for edge cases
- *Mutation tests* — systematic input perturbation
- *Adversarial tests* — handcrafted inputs targeting parser limits
- *Round-trip tests* — parse → dump → parse cycle validation

3.6. Zero-Axiom Policy

The project uses zero axioms beyond Lean's built-in foundations (`propext`, `Quot.sound`, `Classical.choice`). No `sorry` appears anywhere in the codebase. No `partial def` is used — every function has a kernel-checked termination proof.

This means the formal guarantees are as strong as the Lean kernel itself: if the kernel accepts the proofs, the properties hold.

lean4-yaml-verified Test Coverage

YAML 1.2.2 verified parser · Lean 4

yaml-test-suite Compliance

```
Total unique tests:      406
Applicable (YAML 1.2.2): 358 (48 skipped – YAML 1.3 specific)
Passed:                  263
Failed:                   0
Expected fail:           95
Unexpected pass:          0
Timeout:                  0

Correct: 358/358 (100.0%)
```

Reports

[Full Coverage Report](#)

All 406 unique test cases with filtering & sorting

Coverage by Stage

[scalar](#)

58/58 correct (100.0%) · 24 YAML 1.3 skipped

[flow](#)

46/46 correct (100.0%)

[block](#)

99/99 correct (100.0%)

[document](#)

17/17 correct (100.0%) · 7 YAML 1.3 skipped

[advanced](#)

64/64 correct (100.0%) · 17 YAML 1.3 skipped

[error](#)

74/74 correct (100.0%)

Suite Runner (Progressive Stages)

✓ Suite Runner: 869/869 (100.0%)

Each stage tests its own tests plus all lower stages (cumulative). Error tests excluded.

Stage	Total	Passed	Failed	Skipped
scalar	82	58	0	24
flow	128	104	0	24
block	227	203	0	24
document	251	220	0	31
advanced	332	284	0	48
Total	1020	869	0	151

Verified Tests

✓ Internal Verified Test Suites (1232/1232)	
Unit Tests (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/Main.lean)	✓ 10/10
Explicit Key Tests (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ExplicitKeyTests.lean)	✓ 149/149
Flow Completeness Tests (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/FlowTests.lean)	✓ 88/88
Structural Validation Tests (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ValidationTests.lean)	✓ 84/84
Dump Round-Trip Tests (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/DumpRoundTrip.lean)	✓ 117/117
Raw Parse / Compose Tests (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/RawParseTests.lean)	✓ 29/29
YAML 1.2.2 Spec Examples (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/SpecExamples.lean)	✓ 132/132
Schema ↔ Dump Integration (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/SchemaDump.lean)	✓ 68/68
Scanner & TokenParser Tests (Phase 9) (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ScannerTests.lean)	✓ 32/32
YAML 1.2.2 Spec Examples (Scanner/Parser Pipeline) (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ScannerSpecExamples.lean)	✓ 132/132
Adversarial Grammar-Directed Tests (v0.2.13.1) (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/AdversarialGrammarTests.lean)	✓ 154/154
Mutation Testing on yaml-test-suite (v0.2.13.2) (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/MutationSuiteTests.lean)	✓ 45/45
Property-Based Round-Trip Tests (v0.2.13.3) (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/PropertyTests.lean)	✓ 124/124
Production Coverage Analysis (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/ProductionCoverage.lean)	✓ 18/18
Parser Security Limit Tests (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Tests/LimitTests.lean)	✓ 43/43
Demo (https://github.jpl.nasa.gov/pass/lean4-yaml-verified/blob/main/Demo.lean)	✓ 7/7
Total	1232/1232

[Verified Tests Detail](#)

1232/1232 tests across 16 suites

[YAML 1.2.2 Production Coverage](#)

All productions cross-referenced with @[yaml_spec] annotations

Generated by [yaml-test-suite](https://github.com/yaml/yaml-test-suite) (https://github.com/yaml/yaml-test-suite) runner · Lean 4